

Devharsh Trivedi

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AI/ML Security Engineer – Safeguards, Adversarial ML, and Privacy-Preserving Machine Learning

Professional Summary

Ph.D. cybersecurity researcher and CISSP-certified engineer specializing in AI/ML security, adversarial robustness, and privacy-preserving machine learning. Published author on fully homomorphic encryption for ML, federated split-learning, and supervised log-anomaly detection, with 10+ years across software, security, and research. Authorized to work in the U.S.; open to H-1B sponsorship.

Technical Skills

- **ML Security & Safeguards:** Adversarial ML, prompt-injection defense, LLM red-teaming, jailbreak detection, model evaluation, RLHF safety, content moderation
- **Privacy-Preserving ML:** Fully Homomorphic Encryption (CKKS, BFV, BGV), federated and split learning, differential privacy, secure multi-party computation
- **ML & Infrastructure:** PyTorch, TensorFlow, scikit-learn, Hugging Face, MLflow; Python, C/C++, Go, Docker, Kubernetes, AWS, Azure, GCP, CI/CD
- **Cybersecurity:** Threat modeling, vulnerability management, SIEM (Splunk, Azure Sentinel), penetration testing, incident response, NIST 800-53, ISO 27001

Professional Experience

Bowie State University

Lecturer and AI/Security Researcher (CAE-CD program)

2024 - Present

Maryland, USA

- Lead AI-safeguards and threat-assessment automation research; co-authored a \$500K NSF/SFS grant proposal as Co-PI.
- Mentored SURI scholars on federated learning and homomorphic inference and built Python security-telemetry analytics.

Stevens Institute of Technology

Graduate Research Assistant, Privacy-Preserving ML

2020 - 2024

New Jersey, USA

- Authored SigML and SigML++ (log-anomaly detection under FHE; CSCML 2023, MDPI Cryptography) and released Chiku for FHE inference.
- Built Vaultbox, a tamper-evident analytics platform protecting ML training and inference logs (SciSec 2023).

Telecom SudParis

Visiting Research Scholar

2024

France

- Co-developed SplitML, a unified privacy-preserving architecture for federated split-learning (MDPI Electronics 2026).

Oracle / Philips

Senior Engineer (Applications and Software)

2015 - 2019

India

- Delivered ERP REST/SQL/C++ with security testing; built ML and signal-processing toolchains for MRI QA, cutting manual validation 50% on HIPAA software.

Education & Certifications

- **Ph.D., Computer Science** — Stevens Institute of Technology, Hoboken, NJ (2024). GPA 3.59/4.0
- **M.Tech., Computer Science** — Nirma Institute of Technology, India (2016). GPA 8.21/10
- **B.E., Computer Science** — Gandhinagar Institute of Technology, India (2013). GPA 7.52/10
- **Certifications:** (ISC)² CISSP and CC; Microsoft Azure AZ-900 and AI-900; IBM Watson Analytics

Selected Accomplishments

- SigML++, Vaultbox, Chiku, and SplitML: peer-reviewed systems for privacy-preserving ML and security analytics.
- First Prize, **Red Team CTF 2022** (Trend Micro and ISACA NY Metro); IEEE Senior Member; CISSP, Azure AZ-900/AI-900.
- 15+ peer-reviewed publications; reviewer for IEEE and MDPI venues.